March 22, 2026

The Editor
Classical and Quantum Gravity

Dear Editor,

I am submitting the enclosed manuscript, *The Other Half of Gravity: Torsion and the Two-Chamber Structure of the Tetrad Frame*, for consideration as a Letter in *Classical and Quantum Gravity*. The paper is a formal companion to *The Coordinate-Space Geometry of the Tetrad Frame* (CQG-115428), currently under review at your journal.

That companion letter proved that the tetrad equation produces two bounded tetrahedral regions related by spatial parity: TetA (the identity component) and TetB (the parity component). The present letter asks what becomes of this two-chamber structure at the level of the affine connection.

The central result is a lemma: a torsion-free connection cannot distinguish TetA from TetB. The affine connection carries $K^3 = 64$ components; the 40 Christoffel symbols are built entirely from the metric, which is invariant under the parity exchanging the two chambers, while the 24 torsion components depend on the frame field directly and are therefore the necessary carrier of the two-chamber distinction. The zero-torsion condition collapses that distinction. A second result links the antisymmetric count $24 = D \times 2K$ to a coupling polynomial that organises the standard count structure of the tetrad formalism, and this identity holds exclusively at $K = 4$.

A supplementary note derives the TetB field fraction $1/(x{+}1)$ from the exchange symmetry of the two chambers. The ratio of removed to retained field response is $1/x$, which is algebraically identical to the dark-to-baryon mass ratio of the cold dark matter framework, following from four lines of algebra with no halo model and no free parameters.

The manuscript does not propose a modification of general relativity. All claims are either proved locally, derived from the companion geometry, or cited from standard references (Misner, Thorne and Wheeler; Hehl *et al.* 1976; Cartan 1922). A validation script confirms 52/52 symbolic claims.

Because the present letter builds directly on the geometry established in CQG-115428, I would be grateful if the same referee could be considered for both manuscripts, as this would avoid the need to re-establish the companion result within the present review.

The manuscript has not been submitted elsewhere and is not under consideration at any other journal.

Yours sincerely,

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